Wei-Ting Liu

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Education

2024 – Present **National Taiwan University** — B.B.A., Information Management
Double Major, Trans-disciplinary Bachelor Degree (College of Innovation)
Leadership Program

Work Experience

ABConvert — Reinforcement Learning R&D Intern 2026.07 – 2026.09

– **Designed a dual-loop RL system** that trains agents to optimize Shopify stores, aiming to build an accurate, real-world-calibrated simulator that closes the sim-to-real gap.

NTU iNSIGHT — AI Intern 2026.02 – 2026.07

– **Developed an internal service-design product** and **led AI adoption workshops**, moving 80% of staff from chat-based AI use to building their own website products.

ABConvert — Software Engineer Intern 2026.02 – 2026.06

– **Migrated abconvert.io** (10 pages) from Webflow to a self-hosted Next.js app, reaching a 97 Lighthouse SEO score and enabling custom interactive components.

– Built **internal dashboard** features (experiment research, visualization, analysis) from a cross-team needs survey, eliminating ∼70% of manual metric lookups for Customer Success and the CEO.

ABConvert — Startup Generalist Summer Intern (Product/AI) 2025.07 – 2025.09

– **AI-native generalist**: LLM-assisted workflow to compress spec → implementation loop; shipped Next.js + TypeScript frontend (SSR/ISR, streaming UI) at startup pace.

Research Experience

Multi-Agent RL for Decentralized Epidemic Surveillance | 10.5281/zenodo.21619163 2026.02 – Present

– Built a **CTDE multi-agent RL framework** (MAPPO/QMIX/IPPO/CQL) casting Brazil's 26 states as agents alerting on dengue from 258 weeks of cases and Google Trends, benchmarked against an information-parity rule.

– **Decomposed the agents' systematic loss** on historical replay into reward misalignment (partly fixed by a gated false-alarm penalty) and sample inefficiency; the same agents learn control on action-coupled SIR/SIS environments, localizing the failure to task specification, not RL. Manuscript in preparation; 107-checkpoint archive released.

Age-Mixing Structure in Aggregated Urban Mobility Data | 10.5281/zenodo.22152089 2026.08 – Present

– **Measured age-assortative mixing across 10.2 billion rows** of Seoul's Living Migration release (79 months, 424 dong, 16 age bands), pairing the near-zero reading with a synthetic-world calibration that fixes the detection limit.

– Found the co-arrival matrix within **0.00202 of proportionate mixing** every month — 33–35× below a national contact survey — while the pipeline recovers 92–101% of true assortativity on synthetic controls (independently re-implemented, 436/436 checks).

Projects

olcli — Overleaf CLI (open-source contributor) GitHub

TypeScript, Node.js. Contributed `olcli diff` (content-level diff against remote Overleaf projects; 900+ lines, unit/e2e tests, shared file scanning refactored into a reusable module) and a zip-slip path-traversal fix to a 175-star CLI on npm and Homebrew.

Spotify GuessSong GitHub | Live

Next.js, TypeScript, Spotify Web API. Real-time multiplayer music guessing game with 70,000+ monthly active users; host pastes any Spotify playlist URL and the app runs a round-by-round quiz with live scoring.

Skills

Stack	TypeScript, React, Next.js, PostgreSQL, Tailwind, Python, PyTorch, C++
Tools	REST/GraphQL, Git; Service Design, Design Thinking
Coursework	Data Structures, Web Programming, Algorithms, Operating Systems

Awards & Leadership

2025 **PDAO Competition** (Programming Competition) — 5th overall, 1st in IM Department.

2026.07 – 2027.06 **NTU Information Management Student Association** — Vice President.